

ETHAN QIU

613-882-5083 | ethanqiu@gmail.com | linkedin.com/in/qiu-ethan | github.com/qiuethan

EDUCATION

University of Toronto

Sep 2024 - Expected 2028

B.Sc. Computer Science Specialist (AI & Tech Leadership Focus) — GPA: 3.8/4.0

Relevant coursework: Data Structures and Algorithms, Software Design, Linear Algebra, Calculus

EXPERIENCE

Software Engineer (Co-op)

May 2025 - Aug 2025

General Dynamics Mission Systems-Canada

- Designed and delivered internal **Java** and **Python** tooling adopted by **3 engineering teams**, eliminating **10+ manual steps**.
- Owned end-to-end development of a modular **Python automation framework**, reducing regression runtime by **50%**.
- Merged **10+ production PRs** through rigorous code review, improving test coverage and reliability.

Engineering Director - Industry Collaborations

May 2025 - Present

UTMIST (University of Toronto Machine Intelligence Student Team)

- Led cross-functional development across **4 industry projects**, coordinating **20+ developers** with weekly sprint cycles.
- Architected retrieval and ranking microservices using **LangChain**, **vector databases**, and **FastAPI** for production endpoints.

Project Lead

May 2025 - Present

Blueprint, University of Toronto

- Owned delivery of MADE's digital museum platform (**50,000+ artifacts**), managing **10 developers** and shipping first release in **6 weeks**.
- Designed scalable **REST APIs**, role-based access control, and optimized **PostgreSQL** schemas.
- Built **CI/CD pipelines** and deployment workflows, ensuring operational reliability for internal and partner teams.

PROJECTS

Heimer Academy - League of Legends Data Analytics | Python, FastAPI, React, AWS, PostgreSQL [GitHub]

- Built a scalable cloud-native analytics platform on **AWS**, ingesting match data via APIs and computing metrics across **2,000+ games**.
- Designed distributed ML pipelines using **Amazon SageMaker** and **Bedrock**, executing **200,000+ parallelized inferences**.
- 1st Place Overall - Rift Rewind AWS Hackathon.**

Identity Matrix - Persistent Multiplayer AI Simulation | TypeScript, Node.js, FastAPI, Supabase [GitHub]

- Architected a persistent multiplayer system with human-AI avatar handoff, preserving state across distributed sessions (**1st Place Overall - UofTHacks 2026**).
- Decoupled real-time physics (~6-10Hz) from AI inference (1Hz) via event-driven architecture to prevent blocking.
- Implemented probabilistic agent controller combining LLM reasoning with softmax action selection over memory and proximity.

Orbit - Real-Time Social Intelligence System | React, OpenCV, Whisper, LangChain [GitHub]

- Built a distributed system combining real-time vision, transcription, and LLM reasoning for low-latency contextual awareness.
- Engineered sub-second latency pipelines with persistent memory, indexing **200+ identities** in real time.
- Best Use of Groq and Windsurf - Hack the North 2025.**

Bounce Back - AI Mental Wellness Platform | React Native, Firebase, PyTorch, Docker [GitHub]

- Built a cloud-based AI journaling platform with **fine-tuned BERT models** trained on **10,000+ labeled entries**, serving real-time emotion classification to end users.
- Deployed containerized microservices with **Docker**, implementing **CI/CD**, monitoring, and automated logging for operational reliability at scale.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL

Frameworks: React, FastAPI, Django, Node.js, Express, Next.js

Cloud & Tools: AWS (SageMaker, Bedrock, Lambda), Docker, Git, GitHub Actions, PostgreSQL

Libraries: PyTorch, LangChain, OpenCV, pandas, NumPy